RETAIL CASE STUDY / LEVEL 2

Inventory, Promotions and Product Profitability

DD-RET-002 | Business Introduction | Reporting period 2023-01-01 to 2025-12-31

40	L2	30,000	8
QUESTIONS	COMPLEXITY	FACT RECORDS	TABLES

Business purpose

Connect transaction-level sales with inventory snapshots, suppliers, channels and returns to judge promotion and product profitability.

Company background

Crescent Grocery is a fictional grocery chain operating express stores, supermarkets and hypermarkets across Saudi cities. Finance wants one consistent view of sales, margin, promotions, inventory and, at higher levels, customers, payments, suppliers and budgets. Every record is synthetic.

Your role

You are the retail performance analyst. Convert the raw files into a relational model, reconcile activity with financial measures, and explain which commercial and inventory decisions the evidence supports.

Management priorities

Establish sales, discount and margin baselines. Rank stores, categories, products and promotions. Separate rate effects from mix effects. At higher levels, connect inventory movements, payment outcomes, supplier delivery performance and budgets to trading results.

Data grain

L1 rows are one store-product-day. From L2 each row is one transaction line; three consecutive lines form a basket. Inventory snapshots are month-end per store-product; movements are individual stock events. Never add sales and receipts as if both were revenue.

Working with the model

Use single-direction filtering. Keep FulfillmentDateKey, ReturnDateKey and PromisedDateKey inactive and switch them on inside a measure. PromotionID 1 always means no promotion. Budgets are calibrated from synthetic actuals for variance-analysis practice.

Data available to the analyst

- DimDate: 1,096 rows, key DateKey
- DimStore: 15 rows, key StoreID
- DimProduct: 500 rows, key ProductID
- DimSupplier: 40 rows, key SupplierID

- DimChannel: 3 rows, key ChannelID
- DimPromotion: 24 rows, key PromotionID
- FactInventorySnapshot: 40,500 rows, key InventorySnapshotID
- FactSalesTransactions: 30,000 rows, key TransactionID

Business definitions

- Net Sales = Gross sales less promotional discount.
- Gross Margin = Net sales less cost of goods sold.
- Margin % = Gross margin divided by net sales.
- Discount Rate = Discount divided by gross sales.
- Return Rate = Returned lines divided by lines.
- Stockout Rate = Stockout-flagged lines divided by lines.
- OTIF = Delivered on the promised date and in full.
- Shrinkage = Write-offs plus negative adjustments, at cost.

Working assumptions

- Reporting period 2023-01-01 to 2025-12-31; amounts in SAR. All records are synthetic.
- Work in whichever tool you already use - Power BI, Excel, Python, SQL or DuckDB. The method is what is being practised, not the software.
- Use the supplied relationships. Keep inactive date roles inactive unless a question asks for that timing view.
- Sort tied ranked results by the lowest ID. Exclude zero denominators.
- Ask on the discussion board when a question resists you - select the dataset and question number and post.